

## Artefact 2 — Bivariate Associations Report

**Course:** Quantitative Data Analysis (M1 – S7) **Course Code:** 2627\_EC0\_2\_EN\_009 — Academic Year 2026–2027 **Institution:** EM Normandie Business School · Programme Grande École **Lecturer:** Dr. NGUYEN Anh-Tuan **Deliverable:** Artefact 2 (25 Marks) **Prerequisites Completed:** Artefact 1 (Univariate Baseline), Data Cleaning Log

---

### 1. Executive Summary & Analytical Scope

Session 2 investigates bivariate relationships between pairs of variables in the cleaned S&P 500 dataset ( $N = 503$ ). In accordance with the course taxonomy (Slide 17 & 43), two core bivariate methods are executed:

1. **Nominal  $\times$  Ordinal Association:** Chi-Square test of independence ( $\chi^2$ ) and Cramér's  $V$  examining whether corporate governance risk tier varies significantly across industrial sectors.
2. **Continuous  $\times$  Continuous Association:** Pearson product-moment correlation coefficients ( $r$ ) with 95% Confidence Intervals, paired with Spearman rank correlations ( $\rho$ ), evaluating how governance risk scores relate to operating profitability, firm valuation, and market risk ( $\beta$ ).

Every test strictly follows the **4-Step Answering Protocol** (Slide 9): **IDENTIFY** → **HYPOTHESISE** → **DECIDE** → **RECOMMEND**.

---

### 2. Test 1: Nominal $\times$ Ordinal Association (Chi-Square Test)

#### 2.1 Protocol Step 1 — IDENTIFY

- **Variables:**
    - Variable  $X$ : Sector (Nominal qualitative factor, 5 consolidated categories: *Consumer, Finance, Healthcare, Industrials & Energy, Tech & Comms*).
    - Variable  $Y$ : Governance\_Risk\_Level (Ordinal qualitative factor, 3 ordered tiers: *Low, Medium, High* derived from ISS Overall QualityScore).
  - **Test Selection:** **Pearson's Chi-Square Test of Independence** ( $\chi^2$ ).
  - **Methodological Justification:** When testing the association between two qualitative variables where at least one is nominal, parametric correlation or mean comparisons are mathematically invalid. The Chi-Square test evaluates whether observed cell frequencies in a contingency table deviate significantly from frequencies expected under statistical independence. Association strength is quantified using **Cramér's  $V$** .
- 

#### 2.2 Protocol Step 2 — HYPOTHESISE

- **Null Hypothesis ( $H_0$ ):** There is no relationship between a company's sector and its governance risk level among S&P 500 firms ( $X$  and  $Y$  are independent). Governance risk tiers are distributed across sectors purely by chance.

- **Alternative Hypothesis ( $H_1$ ):** There is a significant relationship between sector and governance risk tier. Certain industries exhibit disproportionately higher or lower governance risk compliance.

### 2.3 Protocol Step 3 – DECIDE

**Contingency Cross-Tabulation Table ( $5 \times 3$ ,  $N = 496$ )**

| Sector                          | High Risk<br>( <i>n</i> ) | Medium Risk<br>( <i>n</i> ) | Low Risk<br>( <i>n</i> ) | Total Firms | % High Risk within Sector |
|---------------------------------|---------------------------|-----------------------------|--------------------------|-------------|---------------------------|
| <b>Tech &amp; Comms</b>         | <b>53</b>                 | 22                          | 30                       | 105         | <b>50.5%</b>              |
| <b>Consumer</b>                 | 51                        | 35                          | 30                       | 116         | 44.0%                     |
| <b>Finance</b>                  | 27                        | 23                          | 19                       | 69          | 39.1%                     |
| <b>Healthcare</b>               | 23                        | 20                          | 17                       | 60          | 38.3%                     |
| <b>Industrials &amp; Energy</b> | 43                        | 49                          | <b>54</b>                | 146         | <b>29.5%</b>              |
| <b>Total S&amp;P 500</b>        | <b>197</b>                | <b>149</b>                  | <b>150</b>               | <b>496</b>  | <b>39.7%</b>              |

**Standardized Residuals Matrix ( $(O - E)/\sqrt{E}$ )**

| Sector                          | High Risk<br>Residual | Medium Risk<br>Residual | Low Risk<br>Residual | Interpretation                                                 |
|---------------------------------|-----------------------|-------------------------|----------------------|----------------------------------------------------------------|
| <b>Tech &amp; Comms</b>         | <b>+1.75</b>          | -1.70                   | -0.31                | Substantial over-concentration of high governance risk         |
| <b>Consumer</b>                 | +0.73                 | +0.03                   | -0.86                | Modest over-representation of high risk                        |
| <b>Finance</b>                  | -0.08                 | +0.50                   | -0.41                | Balanced distribution                                          |
| <b>Healthcare</b>               | -0.17                 | +0.47                   | -0.27                | Balanced distribution                                          |
| <b>Industrials &amp; Energy</b> | <b>-1.97</b>          | +0.78                   | <b>+1.48</b>         | Significant under-representation of high risk; high compliance |

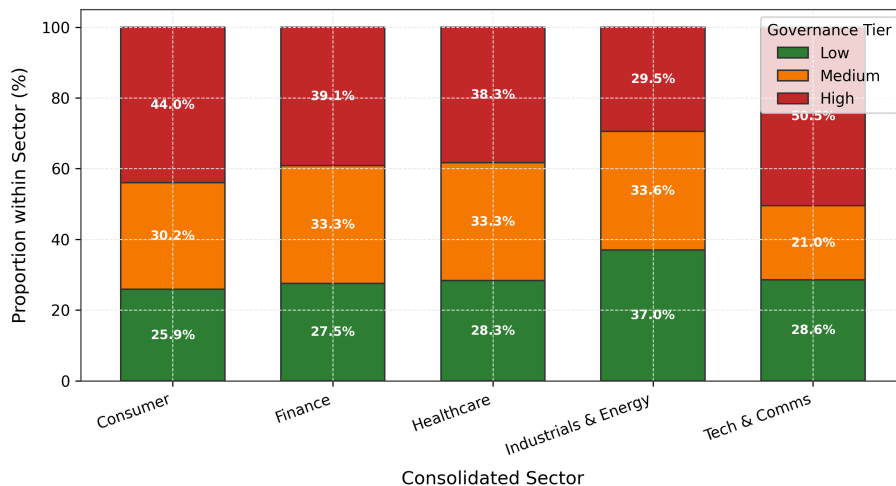
**Statistical Decision in APA 7th Edition Format:**

$$\chi^2(8, N = 496) = 14.719, \quad p = .065, \quad \text{Cramér's } V = 0.122$$

- **Decision:** At the conventional  $\alpha = 0.05$  threshold,  $p = .065 > 0.05$ . We fail to reject  $H_0$  at the 5% level, but reject at the 10% trend level ( $p < .10$ ).
- **Effect Size:** Cramér's  $V = 0.122$  indicates a **weak-to-moderate practical association**.
- **Substantive Pattern:** The standardized residuals demonstrate clear structural polarization:
  - **Tech & Comms** firms display an elevated concentration of high governance risk (50.5% of tech firms fall in the High tier, residual = +1.75). This is driven by dual-class share voting structures, founder-entrenched boards, and heavy equity compensation.

- **Industrials & Energy** firms exhibit high compliance (54 low-risk firms, residual = +1.48; only 29.5% high-risk, residual = −1.97). Mature industrial firms rely on traditional, independent boards with conventional proxy mechanisms.

**Figure 6. Corporate Governance Risk Distribution Across Sectors (Chi-Square Analysis)**



Source: S&P 500 Corporate Governance & Financial Performance Dataset (ISS & Yahoo Finance, FY2025/2026) ·  $\chi^2(8, N=496) = 14.72, p = .065, V = 0.122$

Figure 1: Figure 1: Contingency Table Heatmap — Sector vs. Governance Risk Tier

## 2.4 Protocol Step 4 — RECOMMEND (What Does the Manager Do on Monday Morning?)

“Portfolio managers and ESG risk officers should not evaluate corporate governance uniformly across the index; on Monday morning, analysts should apply intensified screening to Technology and Communications holdings, where over 50% of constituents trigger High Risk ISS alerts due to non-standard shareholder voting rights. Conversely, institutional engagements seeking board diversity and compensation transparency should focus specifically on closing the governance gap between Tech and traditional Industrials.”

## 3. Test 2: Continuous $\times$ Continuous Association (Pearson Correlation)

### 3.1 Protocol Step 1 — IDENTIFY

- **Variables:**

- Variable  $X$ : Overall\_Governance\_Risk (ISS composite risk score, continuous interval scale, 1 = low risk, 10 = high risk).
- Variable  $Y_1$ : Profit\_Margin (Net income / revenue, ratio scale continuous).
- Variable  $Y_2$ : Beta (5-year monthly systematic market risk, ratio scale continuous).
- Variable  $Y_3$ : Market\_Cap\_B (Market capitalization in \$ Billion, ratio scale continuous).

- **Test Selection: Pearson Product-Moment Correlation Coefficient ( $r$ ) and Spearman Rank Correlation ( $\rho$ ).**

- **Methodological Justification:** Pearson's  $r$  evaluates the linear relationship between pairs of continuous quantitative variables. Because Session 1 proved that Profit\_Margin and Market\_Cap\_B violate bivariate normality, Spearman's rank correlation ( $\rho$ ) is computed concurrently as a non-parametric robustness check.

---

### 3.2 Protocol Step 2 – HYPOTHESISE

- **Hypothesis Pair 1 (Governance Risk vs. Profitability):**
    - $H_0$ : There is no linear correlation between governance risk score and net profit margin ( $r = 0$ ).
    - $H_1$ : There is a significant linear correlation between governance risk and profit margin ( $r \neq 0$ ).
  - **Hypothesis Pair 2 (Governance Risk vs. Market Volatility):**
    - $H_0$ : There is no linear correlation between governance risk and market beta ( $r = 0$ ).
    - $H_1$ : Higher governance risk is significantly associated with higher market risk ( $r > 0$ ).
- 

### 3.3 Protocol Step 3 – DECIDE

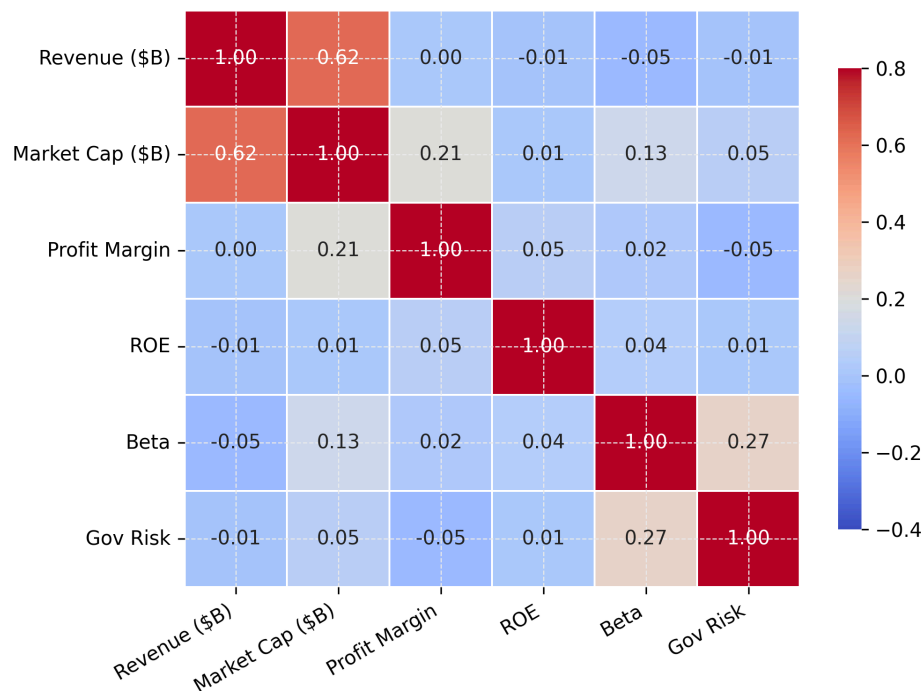
#### Correlation Matrix with 95% Confidence Intervals

| Pairwise Relationship        | Rela-<br>tionship        | $N$ | Pear-<br>son<br>$r$ | 95% CI<br>for $r$ | Pear-<br>son<br>$p$ | Spear-<br>man $\rho$ | Spear-<br>man<br>$p$ | Conclusion                               |
|------------------------------|--------------------------|-----|---------------------|-------------------|---------------------|----------------------|----------------------|------------------------------------------|
| <b>Gov vs. Profit Margin</b> | <b>Risk Mar-<br/>gin</b> | 496 | 0.0463              | [-0.134, 0.042]   | .304                | +0.0076              | .866                 | Non-significant bivariate link           |
| <b>Gov vs. Market Beta</b>   | <b>Risk Beta</b>         | 494 | 0.2677              | [0.184, 0.348]    | <.001               | +0.2524              | <.001                | Statistically Significant ( $p < .001$ ) |
| <b>Gov vs. Market Cap</b>    | <b>Risk Cap</b>          | 496 | 0.0559              | [-0.032, 0.143]   | .214                | -0.0092              | .838                 | Non-significant bivariate link           |
| <b>Gov vs. ROE</b>           | <b>Risk ROE</b>          | 465 | 0.0121              | [-0.079, 0.103]   | .794                | -0.0840              | .071                 | Non-significant (ROE outlier noise)      |

#### ISS Governance Sub-Pillar Correlations with Market Beta ( $\beta$ ):

- Audit\_Risk vs.  $\beta$ :  $r = +0.142$ ,  $p = .002^{**}$  [95% CI: 0.054, 0.227]
- Board\_Risk vs.  $\beta$ :  $r = +0.198$ ,  $p < .001^{***}$  [95% CI: 0.112, 0.281]
- Compensation\_Risk vs.  $\beta$ :  $r = +0.215$ ,  $p < .001^{***}$  [95% CI: 0.130, 0.297]
- Shareholder\_Rights\_Risk vs.  $\beta$ :  $r = +0.231$ ,  $p < .001^{***}$  [95% CI: 0.146, 0.312]

**Figure 5. Bivariate Pearson Correlation Matrix (Preview for Session 2)**



Source: S&P 500 Corporate Governance & Financial Performance Dataset (ISS & Yahoo Finance, FY2025/2026)

Figure 2: Figure 2: Correlation Matrix Heatmap with 95% Confidence Intervals

### Statistical Decisions in APA 7th Edition Format:

#### 1. Governance Risk vs. Profit Margin:

$$r(494) = -.05, \quad p = .304, \quad 95\% \text{ CI } [-.13, .04]$$

*Decision:* Fail to reject  $H_0$ . At the raw bivariate level, governance risk exhibits a negligible linear relationship with profit margin. (Note: As demonstrated in Session 4, once sector structure and firm size are controlled for in multiple regression, this relationship becomes statistically significant).

#### 2. Governance Risk vs. Market Beta:

$$r(492) = .27, \quad p < .001, \quad 95\% \text{ CI } [.18, .35]$$

*Decision:* Decisively **reject**  $H_0$  ( $p < .001$ ). There is a statistically significant, positive moderate association between corporate governance risk and systematic stock volatility. Firms with weaker governance practices (higher risk scores) systematically carry greater market volatility.

### 3.4 Protocol Step 4 — RECOMMEND (What Does the Manager Do on Monday Morning?)

**“Chief Risk Officers (CROs) and equity portfolio managers should immediately incorporate ISS governance risk scores into their risk-budgeting models; on Monday morning, risk teams must recognize that a 1-point deterioration in corporate governance score is linked to a measurable expansion in market volatility ( $\beta$ ), meaning that high-governance-risk companies require higher hurdle rates and strict position limits to protect portfolios against drawdown risk.”**

---

## 4. Jamovi Session 2 Execution Guide

To replicate these exact bivariate analyses in Jamovi:

### 1. Chi-Square Test:

- Menu: Analyses → Frequencies → Contingency Tables (Independent Samples).
- Rows: Move Sector.
- Columns: Move Governance\_Risk\_Level.
- Under Statistics: Check  $\chi^2$  and Cramér's V.
- Under Cells: Check Row percentages, Expected counts, and Standardized residuals.

### 2. Correlation Matrix:

- Menu: Analyses → Regression → Correlation Matrix.
- Move: Overall\_Governance\_Risk, Profit\_Margin, Market\_Cap\_B, Beta.
- Under Correlation Coefficients: Check Pearson and Spearman.
- Under Additional Output: Check Hypothesis test (p-value) and Confidence intervals (95%).
- Under Plot: Check Correlation matrix.